

Render-Timelapse

Record your Blender editing process and create a beautiful video of it.



Digital Artists

Reflect on creation process & showcase on social media

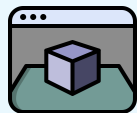


Tutors

On the fly creation of step-by-step visualization for teaching others



Add-on



Rendered Output

No Blender UI. Stylized rendering of primitives



Action Filtering

Skip experimental & faulty steps via undotree



Smart Camera

Smoothen users viewport path to display what is edited



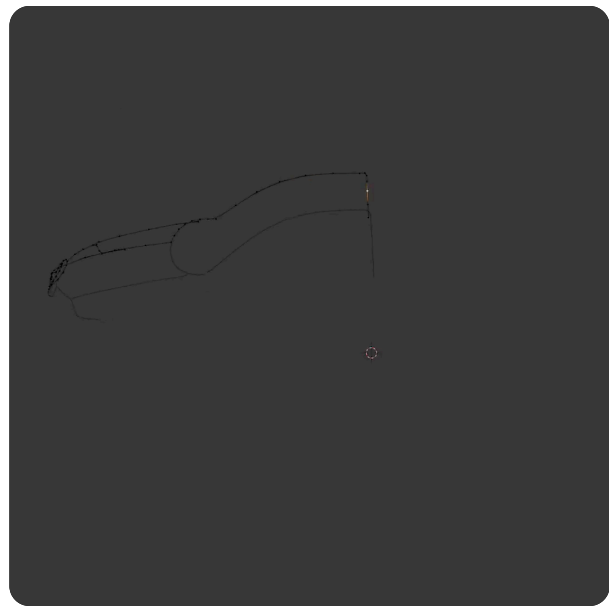
Existing Work: Timelapse Recording

Capture editor/viewport with fixed/rotating camera after some user actions.

- **ScreenSnap Turntable** (designersoup.itch.io)
- **Timelapse Extension** (extensions.blender.org)
- **ReelFlow-Pro** (superhivemarket.com)
- **Eastape Timelapse Pro** (eastapefilms.com)
- **SavePoints** (extensions.blender.org)

Limitations of ScreenSnap

- Captures in-editor shading mode, incl. "Blender Look" for primitives & mode switches
- Captures all actions, even undone actions
- Simplistic camera movement independent of actions



ScreenSnap Example Output

youtube.com/watch?v=iDGZUz5b444

Existing Work



Action Recording

Record sequence of operators to reflect on action or to be repeated again later.

- **Action Recorder** (github.com/InamuraJIN/ActionRecorder)
- **Macros Recorder** (github.com/dairin0d/macros-recorder)
- **Chronicle** (doi.org/10.1145/1866029.1866054)
- **3D Timeline** (doi.org/10.1111/cgf.12311)



Camera Positioning

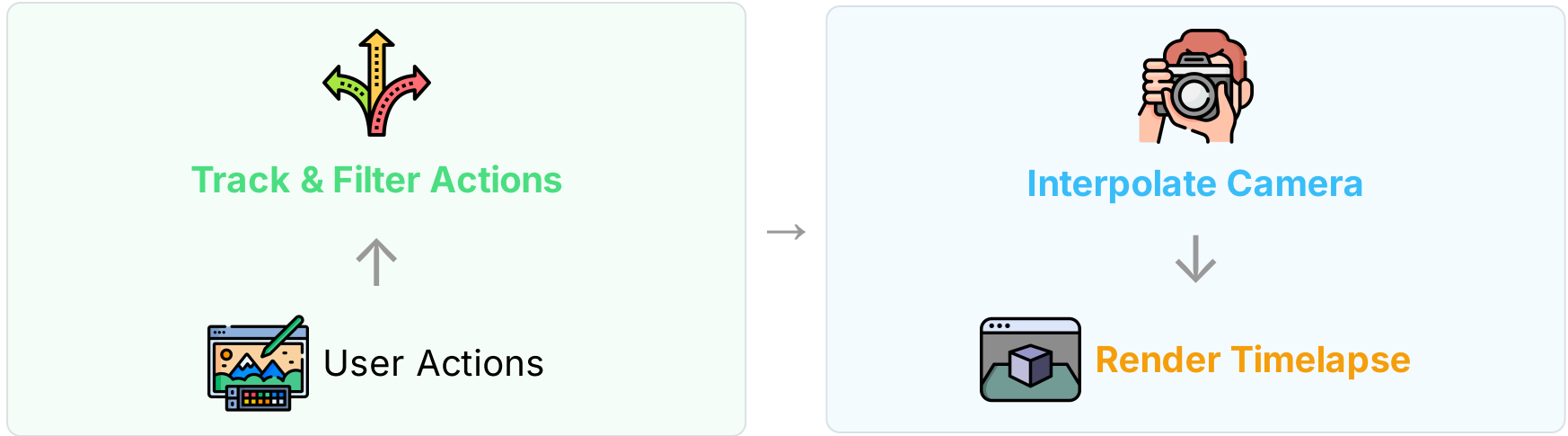
Translate WASD movements into smooth bezier curves.

- **AutoCam** (extensions.blender.org)

Automatic extraction of view-points, -paths, and -surfaces.

- **History Assisted View Authoring** (doi.org/10.1145/2556288.2557009)

User Workflow



User Actions

- 🖥 Viewport
- 📦 Objects
- ✏ Edit Operations
- 🔪 Sculpting
- 🎨 Materials, Shading
- 🔗 Rigging
- 🏗 Modifiers, Geometry Nodes
- ↺ Undo/Redo
- 📄 Scripts, Add-ons
- 📺 *Mode*
- 🎮 *Physics, Particles*
- 📄 *Workspace, UI*
- ▶ *Animation*

Tracking Techniques

Operator-Polling

e.g. `bpy.context.window_manager.operators[i]`

Depsgraph-Update

e.g. `bpy.app.handlers.depsgraph_update_post`

Scene/Data Snapshot

e.g. `bpy.context.active_object`

Viewport Draw

e.g. `bpy.types.SpaceView3D.draw_handler_add`

Undo/Redo

e.g. `bpy.app.handlers.undo_pre`

Rendering

e.g. `bpy.app.handlers.render_complete`

Keylogging

e.g. `modal_handler_add`

Phase 1: Core MVP Functionalities

Jun 17 → Jul 1 (Milestone Presentation)

- ⌚ Track actions (operators, viewport, scene)
- 💾 Recording format & storage
- ↕ Filter via undo-tree
- 📷 Basic camera interpolation
- 🖼️ Render video from recording in new blender project
- ⚙️ UI panel

→ Working add-on with scoped core features and a demo video.

Phase 2: Polish & Stretch Goals

Jul 1 → Jul 15 (Final Presentation & Demo)

- ⚠ Time buffer for unexpected issues
- ✎ Polish UX & edge cases with sensible defaults
- 🕒 Adaptive frame timing
- 📷 Improved camera interpolation
- ⚡ Performance optimizations
- 🖥 Vertex/edge overlays

→ Polished demo video & final presentation ready.

Further Ideas:

Action Inspector

Workflow Analysis

Semantic grouping of actions & scene components

Action animations

Uncertainties



Recording Format:

- JSON/text initially or binary format for less storage
- Representation of undo tree



Camera Interpolation:

- Linear, spline, ML-based approximation of view-path



Scene Types:

- Landscape vs. orbitable
- No viewport change during scripts/geometry nodes



Timing Model:

- Adaptive intervals
- Filter irrelevant actions
- Pauses for camera movement
- Script execution



Performance Hit:

- Add-on overhead running constantly in background
- Render speed for large or long-edited scenes

Q&A

Who already used
a timelapse tool
(for modelling or
elsewhere)?

What would
you use such
an add-on for?

Do you have suggestions for
improvements or further ideas?

What do you think about the additions,
like filtering of undone actions, smart
camera, or custom rendering of
primitives?



github.com/moellh/blender-render-timelapse

